

Quantum Operations and Kraus Representation

Notes from 2023

June 27, 2025

Outline

Motivation: Beyond Unitary Evolution

- ▶ In closed quantum systems, evolution is unitary: $\rho \rightarrow U\rho U^\dagger$. However, real systems interact with environments, undergo measurements, or experience noise, so evolution is more general [citation: 0en.wikipedia.org](https://en.wikipedia.org/wiki/Quantum_operation : : text = In A **quantumoperation** (or **quantumchannel** *) is a mathematical map describing the general transformation of a quantum system dynamics [citation: 1en.wikipedia.org](https://en.wikipedia.org/wiki/Quantum_operation : : text = In
- ▶ Such maps must be **completely positive and trace-preserving (CPTP)** to ensure valid quantum states remain valid [citation: 3quantum.phys.cmu.edu](https://quantum.phys.cmu.edu/QCQI/qitd412.pdf : : text = where intuition : any CPTP map can be realized by coupling the system to an environment, evolving 5quantum.phys.cmu.edu](https://quantum.phys.cmu.edu/QCQI/qitd412.pdf : : text =

Definition: Quantum Operation (CPTP Map)

Definition (Quantum Operation)

A **quantum operation** $\mathcal{E}$ is a linear map on density operators that is (i) trace-preserving and (ii) completely positive. Physically, it represents the most general state evolution (including noise and measurements) [citation : 7quantum.phys.cmu.edu]([https : //quantum.phys.cmu.edu/QCQI/qitd412.pdf](https://quantum.phys.cmu.edu/QCQI/qitd412.pdf) : : text = where

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- Equivalently, by the Stinespring dilation theorem, any quantum operation can be written as

$$\mathcal{E}(\rho) = \text{Tr}_E [U (\rho \otimes |0\rangle\langle 0|) U^\dagger],$$

where U is a unitary on the joint system+environment, and the environment is initialized in a fixed pure state $|0\rangle$

[citation : 9quantum.phys.cmu.edu](https://quantum.phys.cmu.edu/QCQI/qitd412.pdf : : text = 4, In this model, the environment Hilbert space is traced out after the joint

- Properties: Such $\mathcal{E}$ automatically preserves positivity, hermiticity, and trace of ρ [citation : 11quantum.phys.cmu.edu](https://quantum.phys.cmu.edu/QCQI/qitd412.pdf : : text = where